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Temă Geometrie & Alg. Lin 2

Seminar 7:

$$1/ T: \mathbb{R}^3 \rightarrow \mathbb{R}^3, \quad T(x, y, z) = (x+2y, 2y, -2y+z)$$

Am ca T este aplicație liniară ~~de verificare~~

$$T(x+y+z) = T(x) + T(y) + T(z)$$

$$\text{Fie } v_1, v_2 \in \mathbb{R}^3$$

$$\alpha, \beta \in \mathbb{R}$$

$$T(\alpha v_1 + \beta v_2) = \alpha T(v_1) + \beta T(v_2)$$

$$v_1, v_2 \in \mathbb{R}^3 \Rightarrow \begin{cases} v_1 = (x_1, y_1, z_1) \\ v_2 = (x_2, y_2, z_2) \end{cases} \Rightarrow$$

$$\alpha v_1 + \beta v_2 = (\alpha x_1 + \beta x_2, \alpha y_1 + \beta y_2, \alpha z_1 + \beta z_2)$$

$$\begin{aligned} T(\alpha v_1 + \beta v_2) &= \alpha (x_1 + 2y_1, 2y_1, -2y_1 + z_1) + \beta (x_2 + 2y_2, 2y_2, -2y_2 + z_2) \\ &= \alpha T(v_1) + \beta T(v_2) \end{aligned}$$

$\Rightarrow$ Apl. liniară.

$$f: \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

$$f(x, y, z) = (2x - y + 2z, x + 2y - z, x + y + z)$$

Continuous

$$\lambda_1 = 0$$

$$\lambda_2 = 2$$

$$\lambda_3 = 3$$

Subspace

$$S_{\lambda_1} = \begin{cases} 2x - y + 2z = 0 \\ x + 2y - z = 0 \\ x + y + z = 0 \end{cases} \Rightarrow \begin{aligned} 2x - y &= -2z \\ -2z + 2z &= y \\ &\Rightarrow y = 0 \end{aligned}$$

$$\dim = 2$$

$$3x = -3z \Rightarrow x = -z$$

$$\Delta_P = \begin{vmatrix} 2 & -1 \\ -1 & 2 \end{vmatrix} = 3 \neq 0$$

$$\Rightarrow \begin{cases} x, y \text{ nec pr} \\ z = \alpha \text{ nec sec} \end{cases}$$

$$\Rightarrow x = -z$$

$$\Rightarrow V_{\lambda_1} = \{ \lambda(-1, 0, 1) \mid \lambda \in \mathbb{R} \}$$

$$S_{\lambda_2} = \begin{cases} -y + 2z = 0 \\ 2x - y + 2z = 0 \\ -x - z = 0 \\ x + y - z = 0 \end{cases} \Rightarrow \begin{aligned} y &= 2z \\ x &= -z \end{aligned}$$

$$\Delta_P = \begin{vmatrix} 0 & -1 \\ -1 & 0 \end{vmatrix} = 1 \neq 0 \Rightarrow \dim = 2$$

$$\begin{cases} x, y \text{ nec pr} \\ z = \alpha \text{ nec sec} \end{cases} \Rightarrow \begin{aligned} -y &= 2\alpha, y = -2\alpha \\ z &= \alpha \end{aligned}$$

$$V_{\lambda_2} = \{ \alpha(-1, 2, 1) \mid \alpha \in \mathbb{R} \}$$

~~Q1.2.2~~

~~$$\begin{aligned} x_2 + x_3 + 2x_4 + 2x_5 &= 1 \\ x_2 + x_3 + 2x_4 + 2x_5 &= 0 \\ x_2 + x_3 + 2x_4 + 2x_5 &= 0 \end{aligned}$$~~

$$S_{\lambda 3} = \begin{cases} -x - y + 2z = 0 & \Rightarrow x = -y + 2z \Rightarrow x = -y \\ -x - y - z = 0 & \Rightarrow -y - 2z - y - z = 0 \Rightarrow z = 0 \\ x + y - 2z = 0 \end{cases}$$

$$\Delta P = \begin{vmatrix} -1 & 2 \\ -1 & -1 \end{vmatrix} = 3 \neq 0 \Rightarrow r = 2$$

$$\begin{cases} x, z \text{ free} & \Rightarrow x = -y, z = 0 \\ y \text{ free} & \Rightarrow y = 0 \end{cases}$$

$$V_{\lambda 3} = \{ \theta (-1, 1, 0) \mid \theta \in \mathbb{R} \}$$

$$c) m_a(\lambda_1) = m_a(\lambda_2) = m_a(\lambda_3) = 1$$

$$\dim V_{\lambda_1} = m_a(\lambda_1) = 1$$

$$\dim V_{\lambda_2} = m_a(\lambda_2) = 1$$

$$\dim V_{\lambda_3} = m_a(\lambda_3) = 1$$

} $\Rightarrow$ diagonal.

$$d) D = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

Seminara 8:

Căpînoanu Alexandru

$$1/a) A = \begin{pmatrix} 0 & 4 \\ 2 & 2 \end{pmatrix}$$

$$P(\lambda) = \det(A - \lambda I_2) = \begin{vmatrix} 0 - \lambda & 4 \\ 2 & 2 - \lambda \end{vmatrix}$$

$$= -\lambda(2 - \lambda) - 8$$

$$= -2\lambda + \lambda^2 - 8 = \lambda^2 - 2\lambda - 8$$

$$P(\lambda) = 0 \Leftrightarrow \lambda^2 - 2\lambda - 8 = 0$$

$$\Delta = b^2 - 4ac = 4 + 32 = 36$$

$$\lambda_{1,2} = \frac{2 \pm 6}{2} \begin{cases} 4 \\ -2 \end{cases}$$

$$\Rightarrow \begin{cases} \lambda_1 = 4 \\ \lambda_2 = -2 \end{cases}$$

$$m_a(\lambda_1) = m_a(\lambda_2) = 1$$

$$\text{Spec } A = \{-2, 4\}$$

$$S_{\lambda_1} = \begin{cases} -4x + 4y = 0 \\ 2x - 2y = 0 \end{cases} \Rightarrow x = y$$

$$x = \text{vec. pr} \quad (\text{rang} = 1)$$

$$-1 = \text{vec. sec} \Rightarrow x = \alpha$$

$$\Rightarrow V_{\lambda_1} = \{ \alpha(1, 1) \mid \alpha \in \mathbb{R} \}$$

$$S_{\lambda_2} = \begin{cases} 2x + 9 - y = 0 \\ 2x + 4y = 0 \end{cases} \Rightarrow x = -2y$$

$$\begin{cases} x = \text{mec } m \\ y = \beta \text{ mec } \text{soc} \end{cases} \quad (n=1) \Rightarrow x = -2\beta$$

$$\Rightarrow V_{\lambda_2} = \left\{ \beta (-2, 1) \mid \beta \in \mathbb{R} \right\}$$

$$b) A = \begin{pmatrix} 3 & 4 \\ 0 & 5 \end{pmatrix}$$

$$P_{\lambda}(A) = \det(A - \lambda I_2) = \begin{vmatrix} 3-\lambda & 4 \\ 0 & 5-\lambda \end{vmatrix} = (3-\lambda)(5-\lambda)$$

$$P(\lambda) = 0 \Rightarrow (3-\lambda)(5-\lambda) = 0$$

$$\Rightarrow \begin{cases} \lambda_1 = 3 \\ \lambda_2 = 5 \end{cases} \quad \begin{matrix} m_a(\lambda_1) = 1 \\ m_a(\lambda_2) = 1 \end{matrix}$$

$$S_{\lambda_1} = \begin{cases} 0 + 4y = 0 \\ 0 + 2y = 0 \end{cases} \Rightarrow y = 0$$

$$\begin{cases} x = 2 \text{ mec } \text{soc} \\ y = \text{mec } m \end{cases} \quad (n=1)$$

$$V_{\lambda_1} = \left\{ 2(1, 0) \mid 2 \in \mathbb{R} \right\}$$

$$S_{\lambda_2} = \begin{cases} -2 + 4y = 0 \\ 0 + 0 = 0 \end{cases} \Rightarrow x = 2y$$

$$\begin{cases} x = \text{mec } m \\ y = \beta \text{ m. soc} \end{cases} \Rightarrow x = 2\beta$$

$$V_{\lambda_2} = \left\{ \beta (2, 1) \mid \beta \in \mathbb{R} \right\}$$

$$c) A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 2 \\ 0 & 2 & 0 \end{pmatrix}$$

$$P(\lambda) = \det(A - \lambda I_3) = \begin{vmatrix} 1-\lambda & 0 & 0 \\ 0 & -\lambda & 2 \\ 0 & 2 & -\lambda \end{vmatrix}$$

$$= \lambda^2(1-\lambda) - 4 + 4\lambda$$

$$= \lambda^2 - \lambda^3 - 4 + 4\lambda = -\lambda^3 + \lambda^2 + 4\lambda - 4$$

$$P(\lambda) = 0$$

$$\Rightarrow -\lambda^3 + \lambda^2 + 4\lambda - 4 = 0$$

$$-\lambda^2(\lambda - 1) - 4(\lambda - 1) = 0$$

$$= -(\lambda - 1)(\lambda^2 - 4) = -(\lambda - 1)(\lambda - 2)(\lambda + 2) = 0$$

$$\Rightarrow \lambda_1 = 1 \quad m_a(\lambda_1) = 1$$

$$\lambda_2 = 2 \quad m_a(\lambda_2) = 1$$

$$\lambda_3 = -2 \quad m_a(\lambda_3) = 1$$

$$S_{\lambda_1} = \begin{cases} 0 = 0 \\ -\gamma + 2z = 0 \Rightarrow \gamma = 2z \\ 2\gamma - z = 0 \Rightarrow z = 2\gamma \end{cases} \Rightarrow \gamma = z = 0$$

$$\Delta P = \begin{vmatrix} -1 & 2 \\ 2 & -1 \end{vmatrix} = -3 \neq 0 \Rightarrow \text{rg} = 2$$

γ, z free parameters

$x = 0$ free parameter

$$\Rightarrow \gamma = z$$

$$z = \gamma$$

$$\Rightarrow V_{\lambda_3} = \{ \alpha (1, 0, 0) \mid \alpha \in \mathbb{R} \}$$

$$S_{\lambda 2} = \begin{cases} x = 0 \\ -2y + 2z = 0 \\ 2y - 2z = 0 \end{cases} \Rightarrow \begin{cases} x = 0 \\ y = z \end{cases}$$

$$\Delta_P = \begin{vmatrix} -1 & 0 \\ 0 & -2 \end{vmatrix} = 2 \neq 0 \Rightarrow n = 2$$

$$\begin{cases} x, y \text{ nec pr} \\ z = \beta \text{ nec soc} \end{cases} \Rightarrow \begin{cases} x = 0 \\ y = \beta \end{cases}$$

$$V_{\lambda 2} = \{ \beta (0, 1, 1) \mid \beta \in \mathbb{R} \}$$

$$S_{\lambda 3} = \begin{cases} x = 0 \\ 2y + 2z = 0 \\ 2y + 2z = 0 \end{cases} \Rightarrow \begin{cases} x = 0 \\ y = -z \end{cases}$$

$$\Delta_P = \begin{vmatrix} 1 & 0 \\ 0 & 2 \end{vmatrix} \neq 0 \Rightarrow n = 2$$

$$\begin{cases} x, -1 \text{ nec pr} \\ z = \theta \text{ nec soc} \end{cases} \Rightarrow \begin{cases} x = 0 \\ y = \theta \end{cases}$$

$$V_{\lambda 3} = \{ \theta (0, -1, 1) \mid \theta \in \mathbb{R} \}$$

$$d) A = \begin{pmatrix} 0 & 9 & 0 \\ 9 & 0 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

$$P(\lambda) = \det(A - \lambda I_3) = \begin{vmatrix} -\lambda & 9 & 0 \\ 9 & -\lambda & 0 \\ 0 & 0 & 3-\lambda \end{vmatrix}$$

$$= \lambda^2(3-\lambda) - 36(3-\lambda)$$

$$= 3\lambda^2 - \lambda^3 - 108 + 36\lambda$$

$$= -\lambda^2(\lambda - 3) + 36(\lambda - 3)$$

$$= (\lambda - 3)(\lambda^2 - 36) = (\lambda - 3)(\lambda - 6)(\lambda + 6)$$

$$P(\lambda) = 0 \Rightarrow \lambda_1 = 3 \quad m_a(\lambda_1) = 1$$

$$\lambda_2 = 6$$

$$m_a(\lambda_2) = 1$$

$$\lambda_3 = -6$$

$$m_a(\lambda_3) = 1$$

$$S_{\lambda_1} = \begin{cases} -3x + 9y = 0 \\ 9x - 3y = 0 \\ 0 = 0 \end{cases} \Rightarrow \begin{cases} x = 3y \\ x = -3y \end{cases} \Rightarrow y = 0$$

$$\Delta P = \begin{vmatrix} -3 & 9 \\ 9 & -3 \end{vmatrix} = -27 \neq 0 \Rightarrow \eta_g = 2$$

$$\left\{ \begin{array}{l} x, y \text{ nec pr} \\ z = 2 \text{ nec soc} \end{array} \right.$$

$$\Rightarrow x = y = 0$$

$$V_{\lambda_1} = \left\{ \alpha (0, 0, 1) \mid \alpha \in \mathbb{R} \right\}$$

$$S_{\Lambda 2} = \begin{cases} -6x + 9y = 0 \\ 4x - 6y = 0 \\ -3z = 0 \end{cases} \Rightarrow \begin{matrix} z=0 \\ x=y \\ x = \frac{3}{2}y \end{matrix}$$

$$\Delta p = \begin{vmatrix} -6 & 0 \\ 0 & -3 \end{vmatrix} = 18 \neq 0 \Rightarrow \text{rank} = 2$$

$$\begin{cases} x, z \text{ free} \\ y = \beta \end{cases} \Rightarrow \begin{matrix} z=0 \\ x = \frac{3}{2}\beta \end{matrix}$$

$$V_{\Lambda 2} = \left\{ \beta \left(\frac{3}{2}, 1, 0 \right) \mid \beta \in \mathbb{R} \right\}$$

$$S_{\Lambda 3} = \begin{cases} 6x + 9y = 0 \\ 4x + 6y = 0 \\ 9z = 0 \end{cases} \Rightarrow \begin{matrix} z=0 \\ y = -\frac{2}{3}x \end{matrix}$$

$$\Delta p = \begin{vmatrix} 6 & 0 \\ 0 & 9 \end{vmatrix} = 54 \neq 0 \Rightarrow \text{rank} = 2$$

$$\begin{cases} y, z \text{ free} \\ x = 0 \end{cases} \Rightarrow \begin{matrix} z=0 \\ y = -\frac{2}{3}z \end{matrix}$$

$$V_{\Lambda 3} = \left\{ \theta \left(1, -\frac{2}{3}, 0 \right) \mid \theta \in \mathbb{R} \right\}$$

$$2/a) \dim V_{\lambda_1=1} = m_a(\lambda_1) \quad \Rightarrow A. diagonalizabilă$$

$$\dim V_{\lambda_2=1} = m_a(\lambda_2)$$

$$A = \begin{pmatrix} -2 & 0 \\ 0 & 1 \end{pmatrix}$$

b, c, d) analog ↑

$$A_b = \begin{pmatrix} 3 & 0 \\ 0 & 5 \end{pmatrix} \quad A_c = \begin{pmatrix} -2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$A_d = \begin{pmatrix} -6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

$$3/T: \mathbb{R}^3 \rightarrow \mathbb{R}^3, T(x, y, z) = (x+y-1, 2y+3z, y)$$

$$a) T(x) = Ax, x = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \quad A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 2 & 3 \\ 0 & 1 & 0 \end{pmatrix} \in M_3(\mathbb{R})$$

$$\text{Fie } x_1, x_2 = \begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} \text{ și } \alpha_1, \alpha_2 \in \mathbb{R}$$

$$\begin{aligned} T(\alpha_1 x_1 + \alpha_2 x_2) &= A(\alpha_1 x_1 + \alpha_2 x_2) = A(\alpha_1 x_1) + A(\alpha_2 x_2) \\ &= (\alpha_1 A) x_1 + (\alpha_2 A) x_2 = (\alpha_1 A) x_1 + (\alpha_2 A) x_2 \\ &= \alpha_1 (A x_1) + \alpha_2 (A x_2) = \alpha_1 T(x_1) + \alpha_2 T(x_2) \end{aligned}$$

$\Rightarrow T$ este apl. liniară.

$$b) A_T = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 2 & 3 \\ 0 & 1 & 0 \end{pmatrix}$$

$$c) P(\lambda) = \det(A_T - \lambda I_3) = \begin{vmatrix} 1-\lambda & 1 & 0 \\ 0 & 2-\lambda & 3 \\ 0 & 1 & -\lambda \end{vmatrix}$$

$$= (1-\lambda)(2-\lambda)(-\lambda) - 3(1-\lambda)$$

$$= (2-\lambda-2\lambda+\lambda^2)(-\lambda) - 3(1-\lambda)$$

$$= -2\lambda + \lambda^2 + 2\lambda^2 - \lambda^3 - 3 + 3\lambda$$

$$= -\lambda^3 + 3\lambda^2 - \lambda - 3$$

$$= -(\lambda^3 - 3\lambda^2 + \lambda + 3)$$

$$= -(\lambda^2(\lambda - 3) + (\lambda + 3))$$

$$= -(\lambda + 3)(\lambda^2 - 3)$$

$$= -(\lambda + 3)(\lambda - 3)(\lambda + 1)$$

$$= (1-\lambda)(\lambda^2 - 2\lambda - 3)$$

$$= (1-\lambda)(\lambda - 3)(\lambda + 1)$$

$$\Delta = b^2 - 4ac = 4 + 12 = 16$$

$$\lambda_{1,2} = \frac{2 \pm 4}{2} = -1, 3$$

$$\Rightarrow \lambda^2 - 2\lambda - 3 = (\lambda - 3)(\lambda + 1)$$

$$P(\lambda) = 0 \Leftrightarrow \begin{cases} \lambda_1 = 1 & m_a(\lambda_1) = 1 \\ \lambda_2 = 3 & m_a(\lambda_2) = 1 \\ \lambda_3 = -1 & m_a(\lambda_3) = 1 \end{cases}$$

$$= \text{Spec} = \{-1, 1, 3\}$$

$$d) S_{\lambda_1} = \begin{cases} 4x = 0 \Rightarrow x = 0 \\ y + 3z = 0 \\ x - z = 0 \Rightarrow y = z \end{cases} \Rightarrow z = 0$$

$$\text{rg} = 2$$

$$\Delta_P = \begin{vmatrix} 4 & 0 \\ 1 & 3 \end{vmatrix} = 12 \neq 0 \Rightarrow \begin{cases} x, z \text{ nec prime} \\ y = z \text{ nec sec} \end{cases}$$

$$\Rightarrow V_{\lambda_1} = \{ (0, 0, 0) \mid \lambda \in \mathbb{R} \}$$

$$S_{\lambda_2} = \begin{cases} -2x + 4y = 0 \Rightarrow x = 2y \Rightarrow x = 6z \\ -y + 3z = 0 \Rightarrow y = 3z \\ -y - 3z = 0 \end{cases}$$

$$\text{rg} = 2$$

$$\Delta_P = \begin{vmatrix} -2 & 4 \\ 0 & -1 \end{vmatrix} = 2 \neq 0 \Rightarrow \begin{cases} x, y \text{ nec prime} \\ z = \beta \text{ nec sec} \end{cases}$$

$$\Rightarrow \begin{cases} x = 6\beta \\ y = 3\beta \\ z = \beta \end{cases}$$

$$V_{\lambda_2} = \{ \beta (6, 3, 1) \mid \beta \in \mathbb{R} \}$$

$$S_{\lambda_3} = \begin{cases} 2x + 4y = 0 \Rightarrow x = -2y \Rightarrow x = 2z \\ 3y + 3z = 0 \Rightarrow y = -z \\ y + z = 0 \end{cases}$$

$$\text{rg} = 2$$

$$\Delta_P = \begin{vmatrix} 2 & 4 \\ 0 & 3 \end{vmatrix} = 6 \neq 0 \Rightarrow \begin{cases} x, y \text{ nec prime} \\ z = \theta \text{ nec sec} \end{cases} \Rightarrow \begin{cases} x = 2\theta \\ y = -\theta \end{cases}$$

$$\Rightarrow V_{\lambda_3} = \{ \theta (2, -1, 1) \mid \theta \in \mathbb{R} \}$$

$$\dim V_{\lambda_1} = 1 = m_a(\lambda_1)$$

$$\dim V_{\lambda_2} = 1 = m_a(\lambda_2)$$

$$\dim V_{\lambda_3} = 1 = m_a(\lambda_3)$$

$\Rightarrow A$ diagonalizabilă

$$e) B = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

$$B = \{ v_1 = (1, 0, 0), v_2 = (6, 3, 1), v_3 = (2, -1, 1) \}$$

$$C = (v_1 \ v_2 \ v_3) = \begin{pmatrix} 1 & 6 & 2 \\ 0 & 3 & -1 \\ 0 & 1 & 1 \end{pmatrix}$$

$$f: \mathbb{R}^3 \rightarrow \mathbb{R}^3, f(x, y, z) = (x + 3y - z, -3x + 5y - z, 3x + 3y - z)$$

$$a) A_f = \begin{pmatrix} -1 & -3 & 3 \\ 3 & 5 & 3 \\ -1 & -1 & 1 \end{pmatrix}$$

$$b) P(\lambda) = \det(A_f - \lambda I_3) = \begin{vmatrix} -1-\lambda & -3 & 3 \\ 3 & 5-\lambda & 3 \\ -1 & -1 & 1-\lambda \end{vmatrix}$$

$$= (-1-\lambda)(5-\lambda)(1-\lambda) + 3(5-\lambda) + 3(-1-\lambda) + 9$$

$$= (-5 + \lambda - 5\lambda + \lambda^2)(1-\lambda) + 15 - 3\lambda - 3 - 3\lambda + 9 - 9\lambda$$

- 10 -

$$= -x^2(x-2) + 3x(x-2) - 2(x-2)$$

$$= (x-2)(x + 3x - 2) = -(x-2)(x-2)(x-1)$$

$$= -(x-2)^2(x-1)$$

$$P(x)=0 \Leftrightarrow \begin{cases} x_1=1 \\ x_2=2 \end{cases} \in \mathbb{R} \quad \text{Spec}(x) = \{1, 2\}$$

$$m_a(x_1) = 1$$

$$m_a(x_2) = 2$$

$$S_{x_1} = \begin{cases} -2x + 3y - z = 0 \\ -3x + 4y - z = 0 \Rightarrow y = z \\ -3x + 3y = 0 \Rightarrow x = y \end{cases} \Rightarrow x = y = z$$

$$\text{rg} = 2$$

$$\Delta_P = \begin{vmatrix} -2 & 3 \\ -3 & 4 \end{vmatrix} \Rightarrow \begin{cases} x, y \text{ nec } P \\ z = 2 \text{ nec } sec \end{cases}$$

$$\Rightarrow \forall x_1 = \{ \lambda(1, 1, 1) \mid \lambda \in \mathbb{R} \} \Rightarrow x = -1 = z = 2$$

$$S_{x_2} = \begin{cases} 3x + 3y - z = 0 \\ -3x + 3y - z = 0 \\ 3x + 3y - z = 0 \Rightarrow z = -3x + 3y \end{cases}$$

$$\text{rg} = 1$$

$$\begin{cases} x, y \text{ nec } sec \\ z = 0 \end{cases}$$

$$\Rightarrow z = -3\beta + 3\theta$$

$$z = \text{nec } \text{prime}$$

$$V_{\lambda 2} = \{ \beta (1, 0, -3) + \theta (0, 1, 3) \mid \beta, \theta \in \mathbb{R} \}$$

$$c) \dim V_{\lambda 1} = 1 = m_{\lambda}(\lambda_1)$$

$$\dim V_{\lambda 2} = 2 = m_{\lambda}(\lambda_2) \Rightarrow \text{diagonalisierbar}$$

$$d) A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}, \quad B = (v_1 = (1, 1, 1), v_2 = (1, 0, -3), v_3 = (0, 1, 3))$$

$$ii) f: \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

$$f(x, y, z) = (6x - 5y - 3z, 3x - 2y - 2z, 2x - 2y)$$

$$a) A_f = \begin{pmatrix} 6 & -5 & -3 \\ 3 & -2 & -2 \\ 2 & -2 & 0 \end{pmatrix} \in M_3(\mathbb{R})$$

$$b) P(\lambda) = \det(A_f - \lambda I_3) = \begin{vmatrix} 6-\lambda & -5 & -3 \\ 3 & -2-\lambda & -2 \\ 2 & -2 & -\lambda \end{vmatrix}$$

$$\begin{aligned} &= (6-\lambda)(\lambda^2 + 2\lambda + 2) + 15 + 20 - 12 - 6\lambda - 24 + 4\lambda - 1 \\ &= 12\lambda + 6\lambda^2 - 2\lambda^2 - \lambda^3 + 3\lambda - 36 - 2\lambda - 15\lambda \\ &= -\lambda^3 + 4\lambda^2 - 5\lambda + 2 = -\lambda^3 + \lambda^2 + 3\lambda^2 - 3\lambda - 2\lambda + 2 \\ &= -\lambda^2(\lambda-1) + 3\lambda(\lambda-1) - 2(\lambda-1) = -(\lambda-1)^2(\lambda-2) \end{aligned}$$

- 12 -

$$P(x)=0 \Rightarrow \begin{cases} \lambda_1=1 & m_a(\lambda_1)=2 \\ \lambda_2=2 & m_a(\lambda_2)=1 \end{cases}$$

$$\text{Spec} = \{1, 2\}$$

$$\Delta p = \begin{vmatrix} -3 & -2 \\ -2 & -1 \end{vmatrix} = 1 \neq 0$$

$$S_{\lambda_1} = \begin{cases} 5x - 5y - 3z = 0 \\ 3x - 3y - 2z = 0 \\ 2x - 2y - z = 0 \end{cases} \Rightarrow \begin{aligned} 5x &= 3y + 2z \\ 3x &= 2y + z \end{aligned}$$

$$\text{rg} = 2$$

$$\begin{cases} x = 2 \text{ nec. sec} \\ y, z \text{ nec. prime} \end{cases}$$

$$x = y + z$$

$$x = y + z$$

$$y = 2$$

$$z = 0$$

$$V_{\lambda_1} = \{ \lambda (1, 1, 0) \mid \lambda \in \mathbb{R} \}$$

$$S_{\lambda_2} = \begin{cases} 4x - 5y - 3z = 0 \\ 3x - 4y - 2z = 0 \\ 2x - 2y - 2z = 0 \end{cases} \Rightarrow \begin{aligned} 4x - 5y - 3z &= 0 \Rightarrow 3 - 1 = 3z \Rightarrow y = z \\ 3x - 4y - 2z &= 0 \\ 2x - 2y - 2z &= 0 \Rightarrow x - 2y = 0 \Rightarrow x = 2y \end{aligned}$$

$$\Delta p = \begin{vmatrix} 4 & -5 \\ 3 & -4 \end{vmatrix} = -1 \neq 0 \Rightarrow \text{rg} = 2$$

$$\begin{cases} x, y \text{ nec. prime} \\ z = \beta \text{ nec. sec} \end{cases}$$

$$\Rightarrow y = \beta$$

$$x = 2\beta$$

$$V_{\lambda_2} = \{ \beta (2, 1, 1) \mid \beta \in \mathbb{R} \}$$

$$c) \dim V_{\lambda_1} \neq m_a(\lambda_1)=2 \Rightarrow \text{matrix not diag.}$$

Seminar 9:

$$\sqrt{Q} : \mathbb{R}^3 \rightarrow \mathbb{R}$$

$$Q(x) = x_1^2 + 3x_2^2 + x_3^2 - 2x_1x_2 - 4x_2x_3 \quad (\forall) x = (x_1, x_2, x_3) \in \mathbb{R}^3$$

a) Gauss:

$$A = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 3 & -2 \\ 0 & -2 & 1 \end{pmatrix} \quad \text{matr. asoc. f. pătratic Q în raport cu baza canonică}$$

$$\begin{aligned} Q(x) &= (x_1^2 - 2x_1x_2 + x_2^2) + (4x_2^2 - 4x_2x_3 + x_3^2) - 2x_2^2 \\ &= (x_1 - x_2)^2 + (2x_2 - x_3)^2 - 2x_2^2 \end{aligned}$$

$$\begin{cases} y_1 = x_1 - x_2 \\ y_2 = 2x_2 - x_3 \\ y_3 = x_2 \end{cases}$$

$$Q(x) = y_1^2 + y_2^2 - 2y_3^2, \quad (\forall) y = (y_1, y_2, y_3) \in \mathbb{R}^3$$

$$b) A = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 3 & -2 \\ 0 & -2 & 1 \end{pmatrix} \quad \text{matr. asoc. a lui Q}$$

$$\Delta_1 = 1$$

$$\Delta_2 = \begin{vmatrix} 1 & -1 \\ -1 & 3 \end{vmatrix} = 2$$

$$\Delta_3 = \det A = 3 + 0 + 0 - 0 - 4 - 1 = -2$$

$$Q(x) = \frac{1}{\Delta_1} (x_1')^2 + \frac{\Delta_1}{\Delta_2} (x_2')^2 + \frac{\Delta_2}{\Delta_3} (x_3')^2$$

$$= \frac{1}{1} (x_1')^2 + \frac{1}{2} (x_2')^2 - 1(x_3')^2, \quad (x') = (x_1', x_2', x_3') \in \mathbb{R}^3$$

$$2/f: \mathbb{R}^3 \times \mathbb{R}^3 \rightarrow \mathbb{R}$$

$$F(x, y) = 2x_1y_1 + x_2y_2 - 2x_1y_1 - 2x_2y_1 - 2x_2y_3 - 2x_3y_2$$

$$a) \text{ For } x, y, z \in \mathbb{R}^3$$

$$\alpha, \beta \in \mathbb{R}$$

$$x = (x_1, x_2, x_3)$$

$$y = (y_1, y_2, y_3)$$

$$z = (z_1, z_2, z_3)$$

$$F(\alpha x + \beta y, z) = 2(\alpha x_1 + \beta y_1)z_1 + (\alpha x_2 + \beta y_2)z_2 - 2(\alpha x_1 + \beta y_1)z_2 \\ - 2(\alpha x_2 + \beta y_2)z_1 - 2(\alpha x_2 + \beta y_2)z_3 - 2(\alpha x_3 + \beta y_3)z_2$$

$$= \alpha (2x_1z_1 + x_2z_2 - 2x_1z_2 - 2x_2z_1 - 2x_2z_3 - 2x_3z_2) + \\ \beta (2y_1z_1 + y_2z_2 - 2y_1z_2 - 2y_2z_1 - 2y_2z_3 - 2y_3z_2) \\ = \alpha F(x, z) + \beta F(y, z) \quad (1)$$

$$F(x, \alpha y - \beta z) = 2x_1(\alpha y_1 + \beta z_1) + x_2(\alpha y_2 + \beta z_2) - 2x_1(\alpha y_2 + \beta z_2) \\ - 2x_2(\alpha y_1 + \beta z_1) - 2x_2(\alpha y_3 + \beta z_3) - 2x_3(\alpha y_2 + \beta z_2)$$

$$= 2(2x_1y_1 + x_2y_2 - 2x_1y_2 - 2x_2y_1 - 2x_2y_3 - 2x_3y_2) + \beta (2x_1z_1 + \\ x_2z_2 - 2x_1z_2 - 2x_2z_1 - 2x_2z_3 - 2x_3z_2) = \alpha F(x, y) + \beta F(x, z)$$

$$(1) + (2) \Rightarrow F \text{ lineară în ambii arg.}$$

- 2 -

Seminar 10

Cămpineanu Alexandru

$$B = \{f_1 = (1, 1, 1), f_2 = (1, 1, -1), f_3 = (1, -1, -1)\}$$

$$e_1 = \frac{f_1}{\|f_1\|} = \frac{1}{\sqrt{3}} (1, 1, 1)$$

$$e_2 = \frac{e_2'}{\|e_2'\|}, \quad e_2' = f_2 - \sum_{j=1}^1 \langle f_2, e_j \rangle e_j$$

$$\approx (1, 1, -1) -$$

$$= f_2 - \langle f_2, e_1 \rangle e_1$$

$$= (1, 1, -1) - \frac{1}{\sqrt{3}} \cdot \frac{1}{\sqrt{3}} (1, 1, 1)$$

$$= (1, 1, -1) - \frac{1}{3} (1, 1, 1) = \frac{2}{3} (1, 1, -2)$$

$$\Rightarrow \|e_2'\| = \frac{2}{3} \sqrt{6}$$

$$e_2 = \frac{e_2'}{\|e_2'\|} = \frac{\frac{2}{3} \sqrt{6} (1, 1, -2)}{\frac{2}{3} \sqrt{6}} = \frac{1}{\sqrt{6}} (1, 1, -2)$$

$$e_3' = f_3 - \langle f_3, e_1 \rangle e_1 - \langle f_3, e_2 \rangle e_2$$

$$= (1, -1, -1) + \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{3}} (1, 1, 1) - 2 \cdot \frac{1}{\sqrt{6}} \cdot \frac{1}{\sqrt{6}} (1, 1, -2)$$

$$= (1, -1, -1) + \frac{1}{3} (1, 1, 1) - \frac{1}{3} (1, 1, -2) = (1, -1, 0) \Rightarrow \|e_3'\| = \sqrt{2}$$

$$e_3 = \frac{e_3'}{\|e_3'\|} = \frac{1}{\sqrt{2}} (1, -1, 0)$$

$\{e_1, e_2, e_3\} \subset \mathbb{R}^3$ bază ortonormală altă prin P.O.G.S

$$2/B' = \{ e_1 = \frac{1}{\sqrt{2}} (0, 1, 1), e_2 = \frac{1}{\sqrt{6}} (2, -1, 1), e_3 = \frac{1}{\sqrt{3}} (1, 1, -1) \}$$

coord. of $w = (-1, 1, 2)$

$$w = w_1 e_1 + w_2 e_2 + w_3 e_3$$

$$\langle w, e_1 \rangle = w_1 \underbrace{\langle e_1, e_1 \rangle}_1 + w_2 \underbrace{\langle e_2, e_1 \rangle}_0 + w_3 \underbrace{\langle e_3, e_1 \rangle}_0$$

$$= w_1$$

$$\Rightarrow w_1 = \langle w, e_1 \rangle = \frac{3}{\sqrt{2}}$$

$$\langle w, e_2 \rangle$$

analog $w_2 = \langle w, e_2 \rangle = -\frac{1}{\sqrt{6}}$

$$w_3 = \langle w, e_3 \rangle = -\frac{2}{\sqrt{3}}$$